



Decision Making



Decision/Selection statements

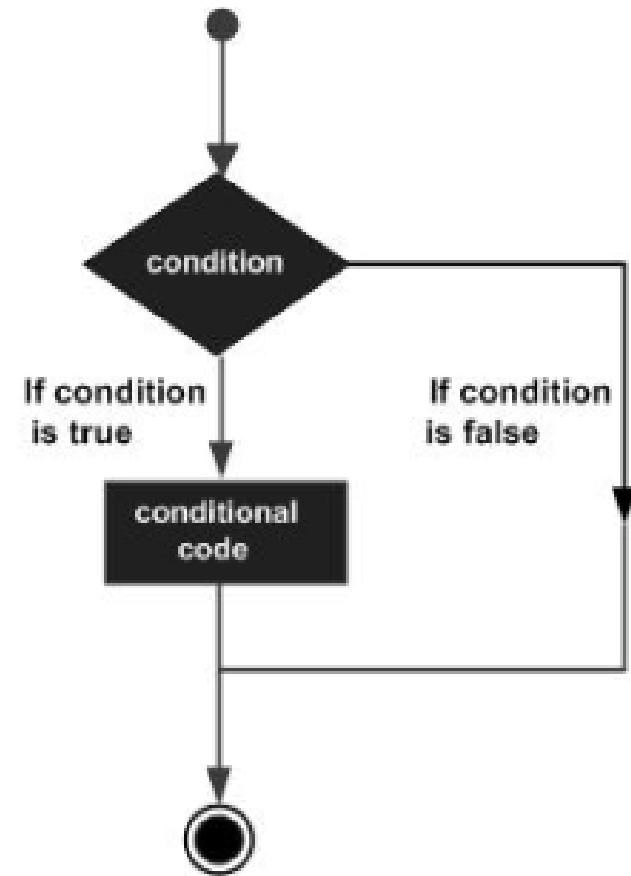
- if
- if else
- nested if
- if else if
- switch case

Simple if

- It is used to decide whether certain statement or block of statements will be executed or not.

Syntax:

```
if (condition)  
{  
  // Executes this block if condition is true  
}
```



```
1 public class IfExample {  
2     public static void main(String[] args) {  
3         int age=20;  
4         if(age>18){  
5             System.out.print("Age is greater than 18");  
6         }  
7     }  
8 }
```

output

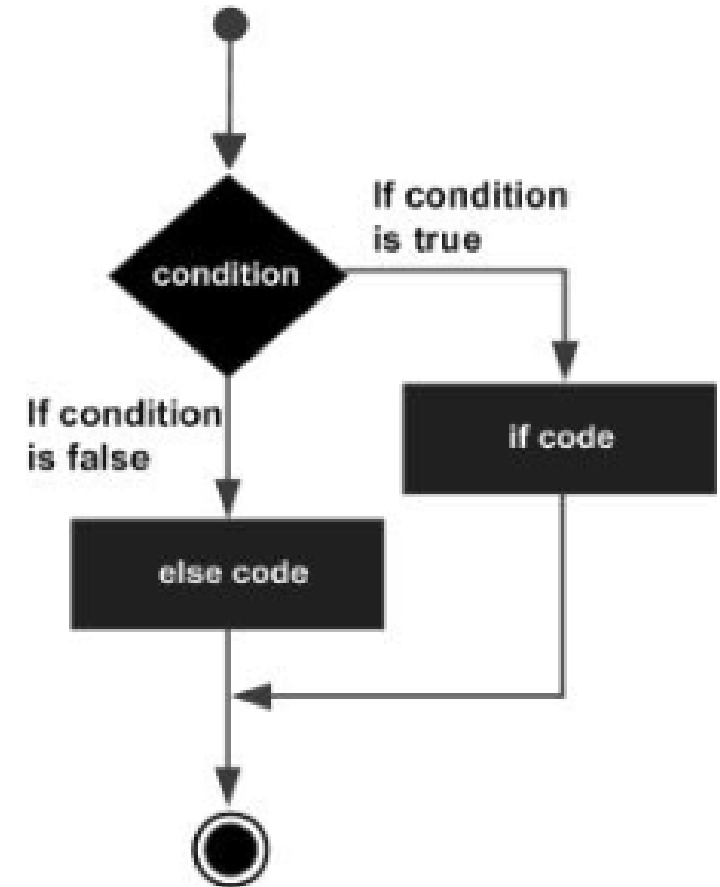
Age is greater than 18

if else

- It is used for two way decision making.
- If the condition is true then the if part is executed and if the condition is false the else part will be executed.

Syntax:

```
if (condition)  
{  
  // Executes this block if condition is true  
}  
else  
{  
  // Executes this block if condition is false  
}
```




```
1 class IfElseDemo
2 {
3     public static void main(String args[])
4     {
5         int i = 10;
6         if (i < 15)
7             System.out.println("i is smaller than 15");
8         else
9             System.out.println("i is greater than 15");
10    }
11 }
```

output

i is smaller than 15

Nested if

- An if else statement can contain any sort of statement within it.
- It can contain another if-else statement
 - An if-else may be nested within the if part.
 - An if-else may be nested within the else part.
 - An if-else may be nested within both parts.

Syntax:

```
if (condition1)  
{  
    // Executes when condition1 is true  
    if (condition2)  
    {  
        // Executes when condition2 is true  
    }  
}
```

```
1 class NestedIfDemo
2 {
3     public static void main(String args[])
4     {
5         int i = 10;
6         if (i == 10)
7         {
8             if (i < 15)
9                 System.out.println("i is smaller than 15");
10            if (i < 12)
11                System.out.println("i is smaller than 12 too");
12            else
13                System.out.println("i is greater than 15");
14        }
15    }
16 }
```

output

i is smaller than 15
i is smaller than 12 too

if else if

- If else if statement is a multiway branch statement.
- In if-else-if statement, as soon as the condition is met, the corresponding set of statements get executed, rest gets ignored.

Syntax:

```
if (condition1)  
    statement1;  
else if (condition2)  
    statement2;  
.  
.  
else  
    statement n;
```

```
1 class ifelseifDemo
2 {
3     public static void main(String args[])
4     {
5         int i = 20;
6
7         if (i == 10)
8             System.out.println("i is 10");
9         else if (i == 15)
10            System.out.println("i is 15");
11        else if (i == 20)
12            System.out.println("i is 20");
13        else
14            System.out.println("i is not present");
15    }
16 }
```

output

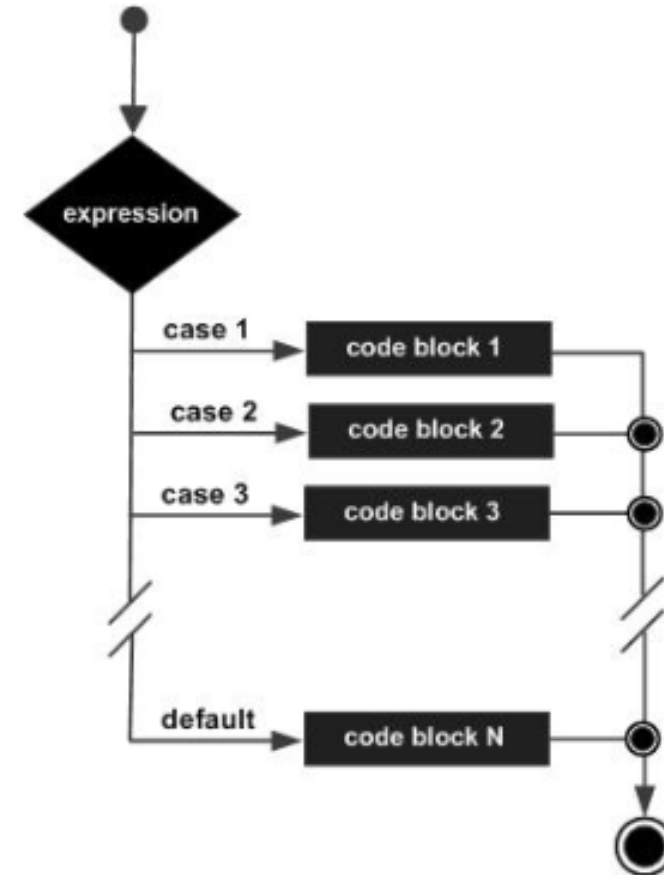
i is 20

switch case

- The switch statement is a multiway branch statement.
- Expression can be of type **byte, short, int char or an enumeration, String.**
- The break statement is optional. If omitted, execution will continue on into the next case.

Syntax:

```
switch (expression)  
{  
    case value1: statement1;  
    break;  
    case value2: statement2;  
    break;  
    .  
    .  
    case valueN: statementN;  
    break;  
    default: statementDefault;  
}
```



```
1 class SwitchCaseDemo
2 {
3     public static void main(String args[])
4     {
5         int i = 9;
6         switch (i)
7         {
8             case 0:
9                 System.out.println("i is zero.");
10                break;
11             case 1:
12                 System.out.println("i is one.");
13                break;
14             case 2:
15                 System.out.println("i is two.");
16                break;
17             default:
18                 System.out.println("i is greater than 2.");
19         }
20     }
21 }
```

output

i is greater than 2.

```
1 // Predict the output
2 class Test {
3     public static void main(String[] args)
4     {
5         int x = 10;
6         if (x) {
7             System.out.println("HELLO");
8         } else {
9             System.out.println("BYE");
10        }
11    }
12 }
```

OUTPUT

1. HELLO
2. Compile time error
3. Runtime Error
4. BYE

```
1 // Predict the output
2 class Test {
3     public static void main(String[] args)
4     {
5         int x = 10;
6         if (x)
7             System.out.println("HELLO");
8         System.out.println("WELCOME");
9
10        else
11        {
12            System.out.println("BYE");
13        }
14    }
15 }
```

OUTPUT

1. HELLO WELCOME
2. HELLO
3. BYE
4. Compile time error

```
1 // Predict the output
2 class Test {
3     public static void main(String[] args)
4     {
5         if (true)
6             ;
7     }
8 }
```

OUTPUT

1. No Output
2. Compile time error
3. Runtime error
4. Runtime Exception


```
1 // Predict the output
2 class MainClass {
3     public static void main(String[] args)
4     {
5         int x = 10;
6         switch (x + 1 + 1) {
7             case 10:
8                 System.out.println("HELLO");
9                 break;
10            case 10 + 1 + 1:
11                System.out.println("BYE");
12                break;
13        }
14    }
15 }
```

OUTPUT

1. Compile time error
2. BYE
3. HELLO
4. No Output

```
1 // Predict the output
2 public class A {
3     public static void main(String[] args)
4     {
5         if (true)
6             break;
7     }
8 }
```

OUTPUT

1. Nothing
2. Error



THANK YOU